Centre Number	Candidate Number	Name

UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
International General Certificate of Secondary Education

CHEMISTRY **0620/02**

Paper 2 (Core)

October/November 2005

1 hour 15 minutes

Candidates answer on the Question Paper.
No Additional Materials required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen in the spaces provided on the Question Paper.
You may use a pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.
The number of marks is given in brackets [ ] at the end of each question or part question.
A copy of the Periodic Table is printed on page 16.

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1	
2	
3	
4	
5	
6	
Total	

This document consists of **16** printed pages.

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1 The diagram shows part of the Periodic Table.

				He
C	N	O	F	Ne
		S	C*l*	Ar
			Br	Kr

(a) Answer these questions using only the elements shown in the diagram.

Write down the symbol for an element which

(i) has five electrons in its outer shell, ... [1]

(ii) has diatomic molecules, ... [1]

(iii) reacts with sodium to form sodium bromide, ... [1]

(iv) is a noble gas, ... [1]

(v) has a giant covalent structure, ... [1]

(vi) has a lower proton number than fluorine, ... [1]

(vii) is the most abundant gas in the air. ..... [1]

(b) Write down a use for each of the following elements.

(i) argon

..... [1]

(ii) helium

..... [1]

(iii) oxygen

..... [1]

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(c) **(i)** Draw a diagram to show the electronic structure of argon.

[2]

(ii) Why is argon very unreactive?

..... [1]

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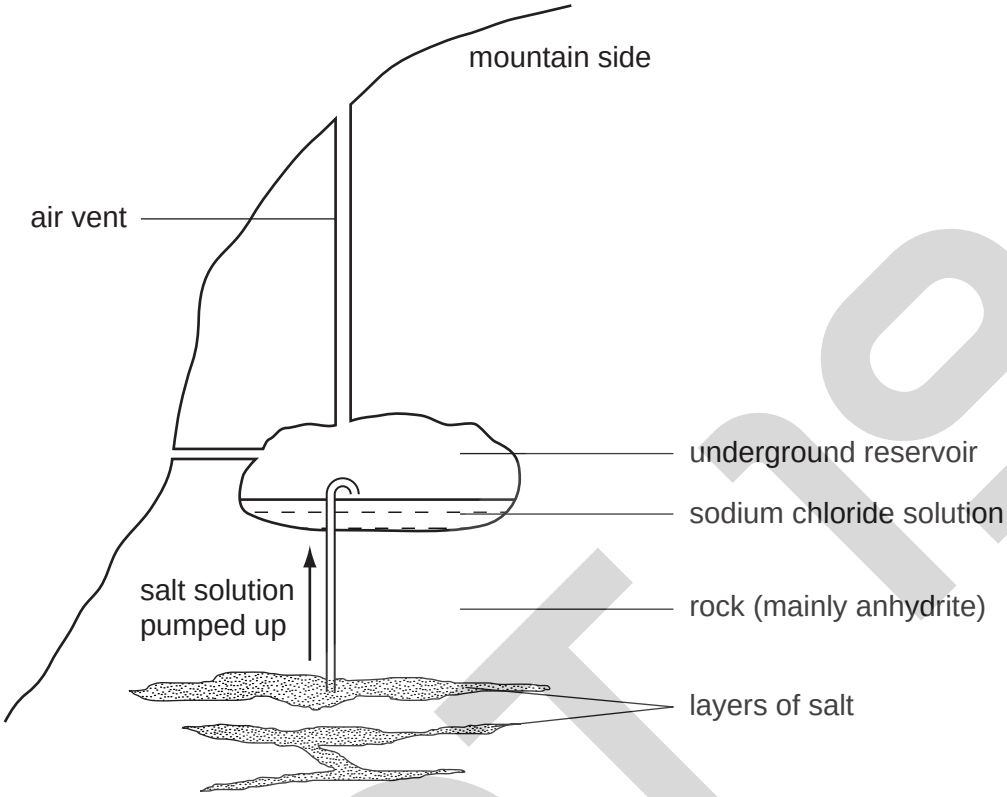
2 The diagram shows the salt mines at Bex in Switzerland.



The salt is dissolved by water from underground springs and then pumped up to a reservoir where it is stored as a solution.

(a) Write the chemical formula for sodium chloride.

..... [1]

(b) Suggest how solid sodium chloride is obtained from the sodium chloride solution.

..... [1]

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(c) Sodium chloride has an ionic giant structure.
Which one of the following best describes an aqueous solution of sodium chloride?
Tick one box.

a mixture of sodium ions and chlorine molecules in water ☐

a mixture of sodium and chlorine atoms in water ☐

a mixture of sodium and chloride ions in water ☐

a mixture of sodium, chloride, oxide and hydrogen ions ☐

[1]

(d) Describe a test for chloride ions.

test .....

result ..... [2]

(e) The rock surrounding the layers of salt is anhydrite.
Pure anhydrite has the chemical formula $CaSO_4$.

(i) State the name of the chemical found in anhydrite.

..... [1]

(ii) Calculate the relative formula mass of the chemical in pure anhydrite.

[1]

(iii) When anhydrite reacts with water, gypsum ($CaSO_4.2H_2O$) is formed.
Complete the equation for this reaction.

$CaSO_4$ + ... $\longrightarrow$ $CaSO_4.2H_2O$ [1]

(iv) Which one of the following describes this reaction?
Put a ring around the correct answer.

combustion fermentation hydration oxidation reduction [1]

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(v) The chemical in anhydrite can be made by reacting calcium hydroxide with sulphuric acid.
Complete the balanced equation for this reaction.

$Ca(OH)_2$ + ... $\longrightarrow$ $CaSO_4$ + ... H_2O [2]

(vi) The spring water running through the rocks changes anhydrite into gypsum.
This reaction is exothermic.
Use this information to explain why the temperature of the mine never falls below 17 °C even in cold winters.

.....

..... [1]

(f) The air inside the mine contains 19% oxygen.
Which one of the following best describes the oxygen level inside the mine compared with that outside the mine?
Tick one box.

the level of oxygen inside the mine is higher ☐

the level of oxygen is the same ☐

the level of oxygen is about a quarter of that of the outside air ☐

the level of oxygen inside the mine is lower ☐

[1]

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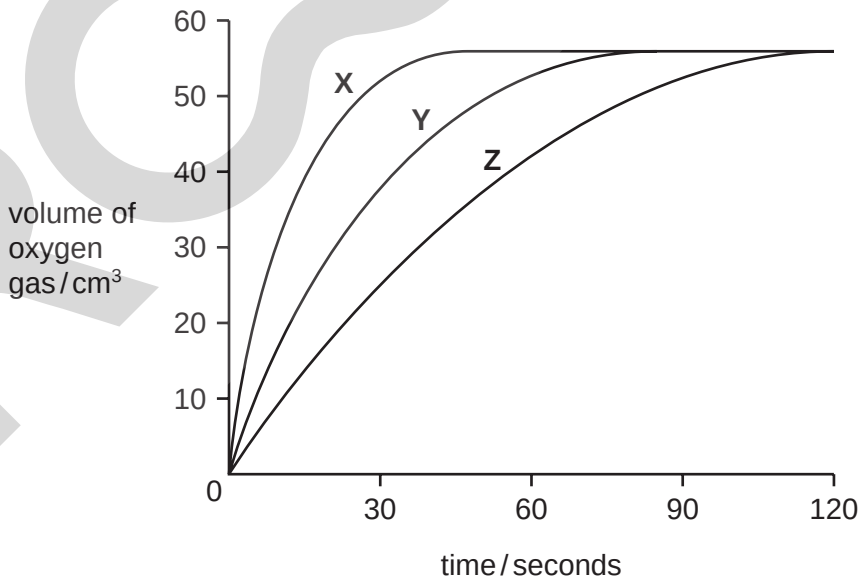
3 Hydrogen peroxide solution, H_2O_2, decomposes slowly in the absence of a catalyst. Oxygen and water are formed.

$$2H_2O_2(aq) \longrightarrow 2H_2O(l) + O_2(g)$$

(a) Draw a diagram of the apparatus you could use to investigate the speed of this reaction.
You must label your diagram.

[3]

(b) Catalyst **X** was added to 50cm^3 of hydrogen peroxide solution at 20°C and the amount of oxygen given off was recorded over a two minute period.
The experiment was repeated with the same amounts of catalyst **Y** and catalyst **Z**.
Apart from the type of catalyst, all conditions were kept the same in the three experiments.
A graph of the results is shown below.



(i) What is a catalyst?

..... [1]

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(ii) Which catalyst, **X**, **Y** or **Z**, produced oxygen gas the fastest?
Explain your answer.

.....

..... [2]

(iii) Why is the final amount of oxygen gas the same in each experiment?

.....

..... [1]

(iv) Many transition metals and their oxides are good catalysts.
State **two** other properties of transition metals which are not shown by other metals.

.....

..... [2]

(c) The experiment with catalyst **Z** was repeated at 40°C.
All other conditions were kept the same.
The speed of the reaction increased.
Explain why, using ideas about particles.

.....

.....

..... [2]

(d) Some enzymes also catalyse the decomposition of hydrogen peroxide.

(i) State one difference between an enzyme and an inorganic catalyst such as a transition metal.

..... [1]

(ii) Enzymes are also responsible for fermentation reactions.
Which one of the following equations **A**, **B**, **C** or **D** describes fermentation?

A $C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O$

B $C_2H_4 + H_2O \longrightarrow C_2H_5OH$

C $C_6H_{12}O_6 \longrightarrow 6C + 6H_2O$

D $C_6H_{12}O_6 \longrightarrow 2C_2H_5OH + 2CO_2$

..... [1]

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4 The list shows some oxides.

calcium oxide
magnesium oxide
nitrogen dioxide
sodium oxide
sulphur dioxide

(a) From this list choose **two** oxides which are basic.
Give a reason for your answer.

.....

..... [2]

(b) **(i)** Which **two** oxides from this list contribute to acid rain?

..... [2]

(ii) How do each of these oxides get into the atmosphere?

name of oxide .....

source of oxide ..... [1]

name of oxide .....

source of oxide ..... [1]

(c) Calcium oxide is manufactured from calcium carbonate.

(i) Complete the word equation for this reaction.

calcium carbonate ⟶ calcium oxide + ... [1]

(ii) What condition is needed for this reaction to take place?

..... [1]

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(d) **(i)** Explain why calcium oxide and sodium oxide cannot be reduced by heating with carbon.

..... [1]

(ii) Copper(II) oxide can be reduced by heating with carbon.
Complete the equation for this reaction.

..... CuO + C ⟶ 2Cu + ... [2]

(iii) What do you understand by the term *reduction*?

.....

..... [1]

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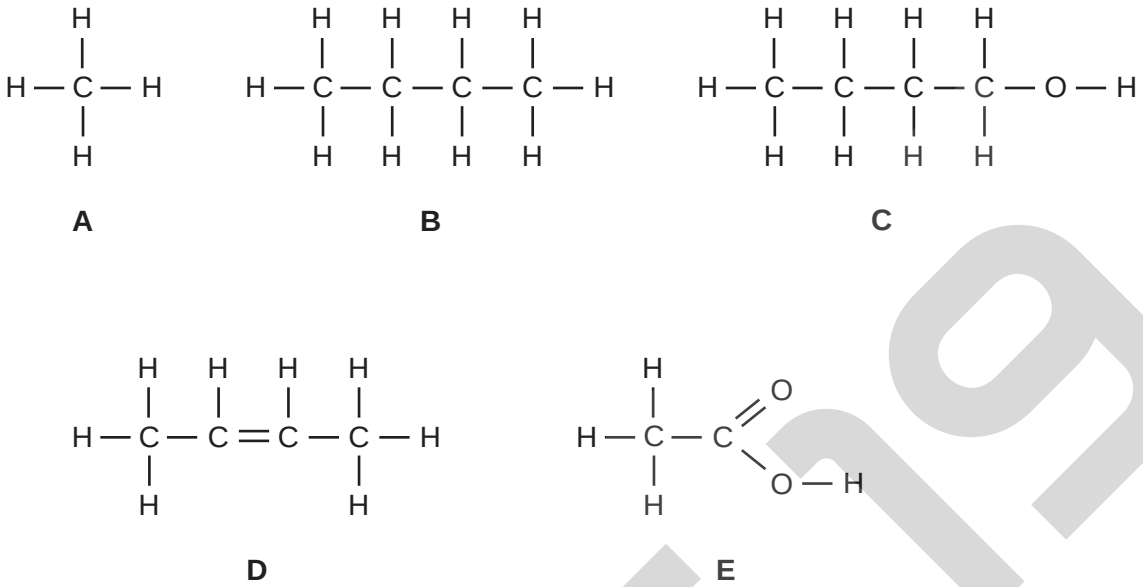
5 The structures of some organic compounds are shown below.



(a) Name compound **A**.

..... [1]

(b) Which **two** of the compounds **A** to **E** belong to the same homologous series?

..... [1]

(c) **(i)** Which one of the compounds **A** to **E** has the same functional group as ethanol?

..... [1]

(ii) Draw the structure of ethanol, showing all atoms and bonds.

[2]

(iii) Describe how ethanol is made in industry from ethene.

.....

..... [2]

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(d) **(i)** Which one of the compounds **A** to **E** is an unsaturated hydrocarbon?

..... [1]

(ii) Describe a chemical test for an unsaturated hydrocarbon.

test .....

result ..... [2]

(e) Compound **E** is acidic.

(i) State the name of compound **E**.

..... [1]

(ii) Describe a test to show that compound **E** is acidic.

test .....

result ..... [2]

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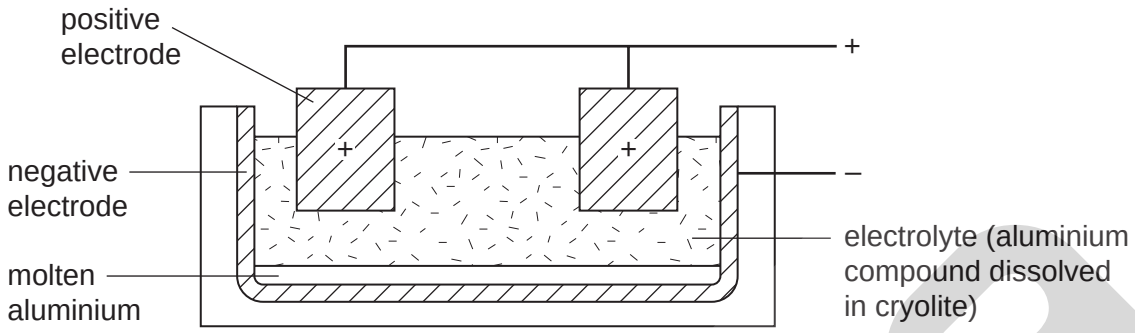
6 The diagram shows an electrolysis cell used to extract aluminium.



(a) What compound of aluminium is used for the electrolyte?

..... [1]

(b) The electrolyte must be molten for the electrolysis to work.
Explain why.

..... [1]

(c) **(i)** State the name of the substance used for the electrodes.

..... [1]

(ii) To which electrode do the aluminium ions move during electrolysis?
Explain your answer.

.....

..... [2]

(d) Complete the following sentences about the molten electrolyte using words from the list below.

bauxite	chemical	cryolite	decreased
electrical	haematite	increased	light

The melting point of the electrolyte is ..... by adding ..... . This means that less ..... energy is needed to melt the electrolyte. [3]

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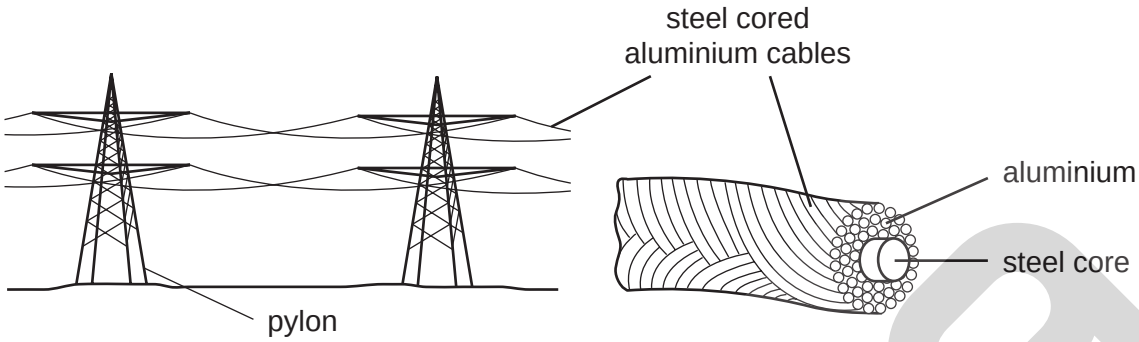
(e) Aluminium is used in overhead power cables.



The table shows some properties of three metals which could be used for the power cables.

metal	relative electrical conductivity	density / grams per cm^3	price / £ per kg	relative strength
aluminium	0.4	2.70	18	9
copper	0.7	8.92	15	30
steel	0.1	7.86	2.7	50

(i) Suggest why aluminium is used for overhead power cables rather than copper.

..... [1]

(ii) Suggest why steel is not used alone for overhead power cables.

..... [1]

(iii) Why is steel used as a core for overhead power cables?

..... [1]

(iv) Electrical insulators are used in parts of the pylons which carry the electrical cables.
Which one of the following is an electrical insulator?
Put a ring around the correct answer.

aluminium ceramic graphite zinc [1]

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(f) Aluminium has many uses.

(i) Why is aluminium used for aircraft bodies?

..... [1]

(ii) Describe a test for aluminium ions.

test .....

result .....

..... [3]

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DATA SHEET
The Periodic Table of the Elements

Group I	Group II											Group III	Group IV	Group V	Group VI	Group VII	Group 0
							1 **H** Hydrogen 1										4 **He** Helium 2
7 **Li** Lithium 3	9 **Be** Beryllium 4											11 **B** Boron 5	12 **C** Carbon 6	14 **N** Nitrogen 7	16 **O** Oxygen 8	19 **F** Fluorine 9	20 **Ne** Neon 10
23 **Na** Sodium 11	24 **Mg** Magnesium 12											27 **Al** Aluminium 13	28 **Si** Silicon 14	31 **P** Phosphorus 15	32 **S** Sulphur 16	35.5 **Cl** Chlorine 17	40 **Ar** Argon 18
39 **K** Potassium 19	40 **Ca** Calcium 20	45 **Sc** Scandium 21	48 **Ti** Titanium 22	51 **V** Vanadium 23	52 **Cr** Chromium 24	55 **Mn** Manganese 25	56 **Fe** Iron 26	59 **Co** Cobalt 27	59 **Ni** Nickel 28	64 **Cu** Copper 29	65 **Zn** Zinc 30	70 **Ga** Gallium 31	73 **Ge** Germanium 32	75 **As** Arsenic 33	79 **Se** Selenium 34	80 **Br** Bromine 35	84 **Kr** Krypton 36
85 **Rb** Rubidium 37	88 **Sr** Strontium 38	89 **Y** Yttrium 39	91 **Zr** Zirconium 40	93 **Nb** Niobium 41	96 **Mo** Molybdenum 42	**Tc** Technetium 43	101 **Ru** Ruthenium 44	103 **Rh** Rhodium 45	106 **Pd** Palladium 46	108 **Ag** Silver 47	112 **Cd** Cadmium 48	115 **In** Indium 49	119 **Sn** Tin 50	122 **Sb** Antimony 51	128 **Te** Tellurium 52	127 **I** Iodine 53	131 **Xe** Xenon 54
133 **Cs** Caesium 55	137 **Ba** Barium 56	139 **La** Lanthanum 57 *	178 **Hf** Hafnium 72	181 **Ta** Tantalum 73	184 **W** Tungsten 74	186 **Re** Rhenium 75	190 **Os** Osmium 76	192 **Ir** Iridium 77	195 **Pt** Platinum 78	197 **Au** Gold 79	201 **Hg** Mercury 80	204 **Tl** Thallium 81	207 **Pb** Lead 82	209 **Bi** Bismuth 83	**Po** Polonium 84	**At** Astatine 85	**Rn** Radon 86
Fr Francium 87	226 **Ra** Radium 88	227 **Ac** Actinium 89															

*58-71 Lanthanoid series
90-103 Actinoid series

140 **Ce** Cerium 58	141 **Pr** Praseodymium 59	144 **Nd** Neodymium 60
Pm Promethium 61	150 **Sm** Samarium 62	152 **Eu** Europium 63
157 **Gd** Gadolinium 64	159 **Tb** Terbium 65	162 **Dy** Dysprosium 66
165 **Ho** Holmium 67	167 **Er** Erbium 68	169 **Tm** Thulium 69
173 **Yb** Ytterbium 70	175 **Lu** Lutetium 71	
232 **Th** Thorium 90	**Pa** Protactinium 91	238 **U** Uranium 92
Np Neptunium 93	**Pu** Plutonium 94	**Am** Americium 95
Cm Curium 96	**Bk** Berkelium 97	**Cf** Californium 98
Es Einsteinium 99	**Fm** Fermium 100	**Md** Mendelevium 101
No Nobelium 102	**Lr** Lawrencium 103	

Key

a
X
b

a = relative atomic mass
X = atomic symbol
b = proton (atomic) number

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

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